Selection of Stem Cell Source

14

Claire Horgan and David Valcárcel

14.1 Introduction

Selection of stem cell source is an important consideration for any physician planning an allogeneic haematopoietic cell transplant (HCT) and has evolved considerably since bone marrow (BM) was used as the stem cell source in the first successful allogeneic HCT in 1968 (Gatti et al. 1968). BM remained the only source of stem cells for the two decades that followed until experimental work demonstrating that peripheral blood (PB) stem cells can be enriched by pretreatment with certain chemotherapy agents and haematopoietic growth factors (Richman et al. 1976; Socinski et al. 1988; Duhrsen et al. 1988) resulted in the first peripheral blood stem cell transplant in 1986 (Korbling and Freireich 2011). Alongside this, the recognition of cord blood (CB) as a rich source of stem cells (Prindull et al. 1978) led to the successful use of cord blood as a third stem cell source in allogeneic HCT in the late 80s (Gluckman et al. 1989).

C. Horgan
Department of Paediatric Bone Marrow Transplant and Cellular Therapy, Royal Manchester Children's Hospital, Manchester University NHS Foundation Trust, Manchester, UK
e-mail: claire.horgan1@nhs.net

D. Valcárcel (✉)
Department of Hematology, Vall d'Hebron Institute of Oncology (VHIO), Vall d'Hebron University Hospital, Universitat Autónoma de Barcelona, Barcelona, Spain
e-mail: dvalcarcel@vhio.net

Following the introduction of PB as a stem cell source, its use evolved significantly, and, by the early 2000s, it had overtaken BM as the predominant stem cell source in adult HCT (Giebel et al. 2019). This trend has continued to increase, with PB being by far the most commonly employed stem cell source, now used in more than 80% of adult allogeneic HCT across the globe (CIBMTR n.d.; Passweg et al. 2023). In contrast, BM remains the stem cell source of choice in the paediatric setting (CIBMTR n.d.), and, despite its appeal as an intuitively attractive alternate donor source in adults and paediatrics (Wynn et al. 2022), cord blood use is declining (Passweg et al. 2021).

14.2 Factors Influencing the Selection of the Source

There are a number of factors that must be considered when selecting the most appropriate stem cell source for HCT and the advantages and disadvantages for both the patient and donor must be considered as well as specific disease-related considerations.

14.2.1 Bone Marrow

The advantages with using BM as a stem cell source include the fact that it can be harvested from a paediatric donor, which is particularly

© The Author(s) 2024
A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_14

relevant in the paediatric matched-related donor setting, and there is no need for donor mobilization using granulocyte colony-stimulating factor (G-CSF). There are also fewer T cells in bone marrow grafts compared with peripheral blood, which results in less graft-versus-host disease (GVHD).

The reduced donor T-cell contamination of the graft may be disadvantageous in the malignant disease setting since T cells are involved in the graft-versus-leukaemia (GVL) effect of allogeneic HCT (Sweeney and Vyas 2019). In addition to this, the cell dose is limited by the volume of the donor marrow and is therefore generally lower than the stem cell dose obtained with mobilized peripheral blood stem cells. Red blood cell (RBC) contamination may be significant, and grafts may therefore require RBC depletion when the donor and recipient are ABO-incompatible, which may further reduce the stem cell dose.

Furthermore, BM harvesting is an invasive procedure requiring a general anaesthetic operation for the donor and, in some circumstances, red blood cell transfusion in those cases in which high cellularity is needed.

14.2.2 Peripheral Blood

The advantages of peripheral blood stem cells include the large donor CD34 stem cell dose, which is generally collected. This helps in more rapid engraftment and immune reconstitution, leading to lower rates of graft failure, reduced risk of infectious complications and faster hospital discharge. Furthermore, the higher number of T cells in the graft may drive engraftment, particularly in reduced intensity conditioning, and may enhance the GVL (graft-versus-leukaemia) effect in malignant diseases. Apheresis of peripheral blood normally includes only very few red blood cells, so depletion is not necessary in most cases.

The disadvantages of peripheral blood stem cells include the increased risk of chronic GVHD due to the higher amount of T-cell content of the product. This may be associated with a prolonged need for immunosuppression therapy in the long-term, post-transplant period and may adversely affect the quality of life. Allogeneic peripheral blood stem cell donation is avoided in the paediatric setting due to ethical considerations with the use of paediatric donor G-CSF mobilization. Additionally, the procedure requires adequate venous access, which can be highly challenging in children.

For donors, PB stem cell harvests are also much less invasive and more tolerable, as no operation is required. However, they need to receive G-CSF to mobilize progenitors (although the safety profile has been demonstrated in large series with long-term follow-up) in some cases, a central vein catheter may be needed, or a second day of collection may be required for some poor mobilizers.

14.2.3 Cord Blood

The advantages of using cord blood include the fact that it is readily available since it is already tissue-typed and cryopreserved. Additionally, the less stringent human leukocyte antigen (HLA) matching required makes it easier to find a donor across HLA barriers (Gragert et al. 2014), which is particularly important for ethnic minorities. Cord blood transplant is associated with rapid immune reconstitution where no serotherapy is used (Chiesa et al. 2012) and higher rates of chimerism in some diseases (Wynn et al. 2022). Recent evidence has also demonstrated that cord blood transplant is associated with reduced relapse and therefore a greater GVL effect (Horgan et al. 2023; Milano et al. 2016).

Disadvantages of cord blood use include a higher transplant-related mortality (Eapen et al. 2007) compared with other donor sources and a

high cost, which makes it challenging in resource-limited settings. The cell dose is also limited, especially in adults or larger patients. This can be overcome using a double cord, although this adds to the complexity of the HCT and further increases the cost and transplant-related mortality (TRM). In addition to this, engraftment may be slower. Recent advances have shown the possibility of expanding progenitor cells from CB, achieving an amount of CD34 near to that obtained in PB transplants (Horwitz et al. 2019). Another disadvantage is the impossibility of obtaining additional cells that can be used for donor lymphocyte infusion (DLI) in the event of disease relapse or the need to improve immune reconstitution.

14.2.3.1 Disease-Related Considerations

Disease-related factors must also be considered when deciding the most appropriate stem cell source for an individual patient. In non-malignant diseases, particularly severe aplastic anaemia, BM is still the preferred stem cell choice due to the lower risk of GVHD (Schrezenmeier et al. 2007). BM is also generally considered to be the stem cell source of choice in the paediatric setting, particularly when paediatric sibling donors are used for the reasons outlined previously. The higher cell dose and more rapid engraftment result in PB being generally preferred as a stem cell source in the adult malignancy setting, particularly where reduced intensity conditioning is used.

14.3 Differences in the Composition of BM vs. PB

BM and PB differ in terms of cellular content, with a higher number 2 to 3 times more $CD34^+$ cells in PB than in BM and a higher number of lymphocytes (in the range of 1 log higher) than in BM. The next table shows the differences in the composition of the different sources.

	Peripheral blood ($\times 10^6$/kg)	Bone marrow ($\times 10^6$/kg)	Ratio of PB to BM
CD34	7.3	2.4	3
CD3	701	49	14
TCR α/β	663	42	14
TCR γ/δ	26	2	13
$CD3^+CD4^+$	393	26	15
$CD4^+CD45RA^+$	188	11	17
$CD4^+CD45RO^+$	169	10	17
$CD3^+CD8^+$	236	18	13
$CD3^+CD4^-CD8^-$	45	5	9
$CD19^+$	93	15	6
$CD16^+CD56^+CD3^-$	77	6	13
$CD45^+CD14^+$	599	25	24

Adapted from Körbling and Ottinger (Körbling and Anderlini 2001)

14.4 Main Results of Studies Comparing BM and PB

There are several randomized trials, retrospective studies and meta-analyses comparing PB and BM as a source of cells in the allogeneic stem cell transplant setting.

Globally, the comparisons suggest that PB transplantation results in faster engraftment and lower graft failure compared to bone marrow (BM) transplantation. However, there is a higher incidence of acute and chronic GVHD associated with PB transplantation.

The risk of relapse appears to be slightly lower when using PB, possibly due to the higher incidence of GVHD. This effect is particularly evident in cases involving HLA identical donors and advanced stage diseases, resulting in a benefit in disease-free survival without significant differences in the overall survival across most studies. The main results of some of these studies are presented in the following table.

	Peripheral blood	Bone marrow	*P*-value
SCTCG meta-an, 1:1. HLA id sib (S.C.T.C Group 2005)	*N* = 538	*N* = 554	
Neutrophil engraftment (days)	14	21	<0.001
Platelet engraftment (days)	14	22	<0.001
Acute GVHD (%)	41	38	0.5
Grade III/IV acute GVHD (%)	26	21	0.03
Chronic GVHD (%)	73	56	0.01
Extensive GVHD (%)	51	35	0.01
3-Year NRM (%)	16	25	0.04
3-Year relapse (%)	21	27	0.01
3-Year DFS (%)	59	53	0.02
3-Year OS (%)	62	59	0.17
Holtick meta-an, Rel and URD (Holtick et al. 2015)	*N* = 653	*N* = 677	
Neutrophil engraftment (days)	16	20	<0.001
Platelet engraftment (days)	13	19	<0.001
Acute GVHD (HR)	Reference	1.03	0.67
Grade III/IV acute GVHD (HR)	Reference	0.75	0.07
Chronic GVHD (HR)	Reference	0.72	<0.001
Extensive GVHD (HR)	Reference	0.69	0.006
NRM (HR)	Reference	0.98	0.91
Relapse (HR)	Reference	1.3	0.07
DFS (HR)	Reference	1.04	0.6
OS (HR)	Reference	1.07	0.43
Anasetti. RCT. MA. UD (Anasetti et al. 2012)	*N* = 273	*N* = 278	
Neutrophil engraftment (days)	Reference	5 days later	0.001
Platelet engraftment (days)	Reference	7 days later	0.001
Engraftment failure (%)	3	9	0.002
Acute GVHD (%)	≈50	≈50	NS
Chronic GVHD (%)	53	41	0.01
Extensive GVHD (%)	48	32	0.001
2-Year NRM (%)	≈25	≈25	0.66
2-Year relapse (%)	≈30	≈30	0.74
2-Year DFS (%)	≈45	≈45	0.38
2-Year OS (%)	≈50	≈50	0.33
Savani. Registry. RIC (Savani et al. 2016)	*N* = 9011	*N* = 837	
Neutrophil engraftment (days)	16	20	<0.001
Acute GVHD (%)	24	17	0.005
Chronic GVHD (%)	36	29	<0.001
2-Year NRM (%)	20	18	0.19
2-Year relapse (%)	35	43	<0.001
2-Year DFS (%)	44	39	0.009
2-Year OS (%)	50	47	0.05
Ruggeri. Registry. Haplo donor (Ruggeri et al. 2018)	*N* = 191	*N* = 260	
Neutrophil engraftment (days)	17	18	0.001
Acute GVHD (%)	38	21	0.01
Grade III/IV acute GVHD (HR)	14	4	0.01
Chronic GVHD (%)	32	36	0.28
2-Year NRM (%)	23	23	0.61
2-Year relapse (%)	22	26	0.38
2-Year DFS (%)	54	49	0.39
2-Year OS (%)	56	55	0.57
2-Year GRFS	43	44	0.82

	Peripheral blood	Bone marrow	*P*-value
Simonin. Registry. Paediatrics. (Simonin et al. 2017)	*N* = 719	*N* = 1743	
Neutrophil engraftment (days)	16	19	0.001
Acute GVHD (odd ratio)	Reference	1.07	0.55
Chronic GVHD (%)	44	21	<0.001
3-Year NRM (%)	20	12	0.002
3-Year relapse (%)	26	29	0.29
3-Year DFS (%)	54	59	<0.001
3-Year OS (%)	62	67	<0.001

Meta-an Meta-analysis, *Sib* Sibling, *Rel* Related, *UR* Unrelated, *NS* Not significant, *MA* Myeloablative, *RIC* Reduced intensity conditioning

14.5 The Role of Cord Blood

Cord blood is an intuitively attractive stem cell source owing to its greater permissiveness for HLA mismatch and subsequent increased donor pool (Schrezenmeier et al. 2007), low rates of chronic GVHD (Keating et al. 2019; Sharma et al. 2020; Barker et al. 2020) and ready availability, yet its recent decline has been mirrored by a rise in haploidentical transplantation (Passweg et al. 2021). This is largely due to the fact that cord blood transplants are more challenging, with a higher TRM (Eapen et al. 2007) that may in part be attributed to lower stem cell doses and T-replete strategies. Despite this, cord remains an important stem cell source and has specific utility in both malignant and non-malignant disease settings. Indeed, T-replete cord blood transplant is associated with reduced relapse and improved event-free survival in paediatric patients with high-risk myeloid malignancy (Horgan et al. 2023), and a similar reduction in relapse with cord has been observed in adults (Milano et al. 2016; Sharma et al. 2020). In non-malignant disease, the superior chimerism associated with cord blood (Church et al. 2007) results in improved outcomes in patients with metabolic disease (Lum et al. 2017; Boelens et al. 2009). Strategies to reduce the high TRM associated with cord blood transplant may serve to broaden its applicability as a stem cell source and safeguard the future of cord blood transplant, particularly in settings where the use of alternate strategies may be associated with an adverse outcome. In this line, a recent randomized study has shown that ex vivo expansion of CB is associated with faster engraftment and less early transplant complications in comparison with standard CB transplantation.

Key Points

- The use of PB compared to BM is associated with earlier haematological recovery, a higher risk of GVHD and, possibly, a greater GVL effect. Overall, there are no significant differences observed in terms of overall survival or disease-free survival.
- In patients at a high risk of relapse, the use of PB may be associated with a GVL effect, especially in the context of reduced-intensity transplantation.
- In general, for patients with non-neoplastic diseases, particularly aplastic anaemia, the use of BM is preferred.
- Since the overall results are similar and there is a lower incidence of chronic GVHD, in patients with standard-risk diseases, the consideration of BM transplantation is important as it may be associated with better long-term quality of life.

References

Anasetti C, Logan BR, Lee SJ, et al. Peripheral-blood stem cells versus bone marrow from unrelated donors. N Engl J Med. 2012;367:1487–96.

Barker JN, Devlin SM, Naputo KA, Skinner K, Maloy MA, Flynn L, et al. High progression-free survival after

intermediate intensity double unit cord blood transplantation in adults. Blood Adv. 2020;4(23):6064–76.
Boelens JJ, Rocha V, Aldenhoven M, Wynn R, O'Meara A, Michel G, et al. Risk factor analysis of outcomes after unrelated cord blood transplantation in patients with hurler syndrome. Biol Blood Marrow Transplant. 2009;15(5):618–25.
Chiesa R, Gilmour K, Qasim W, et al. Omission of in vivo T-cell depletion promotes rapid expansion of naive CD4+ cord blood lymphocytes and restores adaptive immunity within 2 months after unrelated cord blood transplant. Br J Haematol. 2012;156:656–66.
Church H, Tylee K, Cooper A, Thornley M, Mercer J, Wraith E, et al. Biochemical monitoring after haemopoietic stem cell transplant for hurler syndrome (MPSIH): implications for functional outcome after transplant in metabolic disease. Bone Marrow Transplant. 2007;39(4):207–10.
CIBMTR. Current Uses and Outcomes of Hematopoietic Cell Transplantation in the US. 2022 Summary Slides; n.d.
Duhrsen U, Villeval JL, Boyd J, et al. Effects of recombinant human granulocyte colony-stimulating factor on hematopoietic progenitor cells in cancer patients. Blood. 1988;72:2074–81.
Eapen M, Rubinstein P, Zhang MJ, et al. Outcomes of transplantation of unrelated donor umbilical cord blood and bone marrow in children with acute leukaemia: a comparison study. Lancet. 2007;369:1947–54.
Gatti RA, Meuwissen HJ, Allen HD, Hong R, Good RA. Immunological reconstitution of sex-linked lymphopenic immunological deficiency. Lancet. 1968;2:1366–9.
Giebel S, Boumendil A, Labopin M, et al. Trends in the use of hematopoietic stem cell transplantation for adults with acute lymphoblastic leukemia in Europe: a report from the acute leukemia working Party of the European Society for blood and marrow transplantation (EBMT). Ann Hematol. 2019;98:2389–98.
Gluckman E, Broxmeyer HA, Auerbach AD, et al. Hematopoietic reconstitution in a patient with Fanconi's anemia by means of umbilical-cord blood from an HLA-identical sibling. N Engl J Med. 1989;321:1174–8.
Gragert L, Eapen M, Williams E, et al. HLA match likelihoods for hematopoietic stem-cell grafts in the U.S. registry. N Engl J Med. 2014;371:339–48.
Holtick U, Albrecht M, Chemnitz JM, et al. Comparison of bone marrow versus peripheral blood allogeneic hematopoietic stem cell transplantation for hematological malignancies in adults - a systematic review and meta-analysis. Crit Rev Oncol Hematol. 2015;94:179–88.
Horgan C, Mullanfiroze K, Rauthan A, et al. T-replete cord transplants give superior outcomes in high risk and relapsed/refractory paediatric myeloid malignancy. Blood Adv. 2023;7:2155.
Horwitz ME, Wease S, Blackwell B, et al. Phase I/II study of stem-cell transplantation using a single cord blood unit expanded ex vivo with nicotinamide. J Clin Oncol. 2019;37:367–74.
Keating AK, Langenhorst J, Wagner JE, et al. The influence of stem cell source on transplant outcomes for pediatric patients with acute myeloid leukemia. Blood Adv. 2019;3:1118–28.
Körbling M, Anderlini P. Peripheral blood stem cell versus bone marrow allotransplantation: does the source of hematopoietic stem cells matter? B Blood. 2001;98:2900–8.
Korbling M, Freireich EJ. Twenty-five years of peripheral blood stem cell transplantation. Blood. 2011;117:6411–6.
Lum SH, Miller WP, Jones S, Poulton K, Ogden W, Lee H, et al. Changes in the incidence, patterns and outcomes of graft failure following hematopoietic stem cell transplantation for hurler syndrome. Bone Marrow Transplant. 2017;52(6):846–53.
Milano F, Gooley T, Wood B, et al. Cord-blood transplantation in patients with minimal residual disease. N Engl J Med. 2016;375:944–53.
Passweg JR, Baldomero H, Chabannon C, et al. Hematopoietic cell transplantation and cellular therapy survey of the EBMT: monitoring of activities and trends over 30 years. Bone Marrow Transplant. 2021;56:1651–64.
Passweg JR, Baldomero H, Ciceri F, et al. Hematopoietic cell transplantation and cellular therapies in Europe 2021. The second year of the SARS-CoV-2 pandemic. A report from the EBMT activity survey. Bone Marrow Transplant. 2023;58:647–58.
Prindull G, Prindull B, Meulen N. Haematopoietic stem cells (CFUc) in human cord blood. Acta Paediatr Scand. 1978;67:413–6.
Richman CM, Weiner RS, Yankee RA. Increase in circulating stem cells following chemotherapy in man. Blood. 1976;47:1031–9.
Ruggeri A, Labopin M, Bacigalupo A, et al. Bone marrow versus mobilized peripheral blood stem cells in haploidentical transplants using posttransplantation cyclophosphamide. Cancer. 2018;124:1428–37.
S.C.T.C Group. Allogeneic peripheral blood stem-cell compared with bone marrow transplantation in the Management of Hematologic Malignancies: an individual patient data meta-analysis of nine randomized trials. J Clin Oncol. 2005;23:5074–87.
Savani BN, Labopin M, Blaise D, et al. Peripheral blood stem cell graft compared to bone marrow after reduced intensity conditioning regimens for acute leukemia: a report from the ALWP of the EBMT. Haematologica. 2016;101:256–62.
Schrezenmeier H, Passweg JR, Marsh JC, et al. Worse outcome and more chronic GVHD with peripheral blood progenitor cells than bone marrow in HLA-matched sibling donor transplants for young patients with severe acquired aplastic anemia. Blood. 2007;110:1397–400.
Sharma P, Purev E, Haverkos B, et al. Adult cord blood transplant results in comparable overall survival and

improved GRFS vs matched related transplant. Blood Adv. 2020;4:2227–35.
Simonin M, Dalissier A, Labopin M, et al. More chronic GvHD and non-relapse mortality after peripheral blood stem cell compared with bone marrow in hematopoietic transplantation for paediatric acute lymphoblastic leukemia: a retrospective study on behalf of the EBMT Paediatric diseases working party. Bone Marrow Transplant. 2017;52:1071–3.
Socinski MA, Cannistra SA, Elias A, et al. Granulocyte-macrophage colony stimulating factor expands the circulating haemopoietic progenitor cell compartment in man. Lancet. 1988;1:1194–8.
Sweeney C, Vyas P. The graft-versus-leukemia effect in AML. Front Oncol. 2019;9:1217.
Wynn R, Nataraj R, Nadaf R, Poulton K, Logan A. Strategies for success with umbilical cord Haematopoietic stem cell transplantation in children with malignant and non-malignant disease indications. Front Cell Dev Biol. 2022;10:836594.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

